Foundations of LLMs & Cloud Computing Basics

Overview

14 Aug 2026

- **establish foundation in LLM architecture**
  1. **describe the key differences between major LLM providers**
     i. **OpenAI**
     ii. **Anthropic**
     iii. **Google**

- **set up and configure cloud development environments for AI development**
  1. **make your first API calls through the UNC AI Gateway**
     i. **Azure**
     ii. **OpenAI**

- **reinforce understanding of transformers**
  1. **explain the self-attention mechanism and its role in transformer architecture**

Required Readings

- Vaswani, A., et al. (2017). "Attention Is All You Need."
  - the foundational transformer paper
  - focus on Sections 1-3 and the attention mechanism diagrams
  - **https://arxiv.org/abs/1706.03762**

- Brown, T. B., et al. (2020). "Language Models are Few-Shot Learners."
  - the GPT-3 paper that introduced modern prompting
  - read the introduction and emergent abilities section
  - **https://arxiv.org/abs/2005.14165**

- OpenAI (2023). "GPT-4 Technical Report."
  - understand current capabilities and limitations

- https://arxiv.org/abs/2303.08774

Required Videos

- Understanding Transformer Architecture (4 min)
- Understanding Attention Mechanisms (4 min)
- Introduction to Cloud Platforms (3 min)
- Overview of Current Model Landscape (5 min)

Supplementary Videos

- Andrej Karpathy - "Let's build GPT: from scratch" (2 hours)
  - Highly recommended for deep understanding
  - https://www.youtube.com/watch?v=kCc8FmEb1nY
- MIT 6.S191 - "Transformers and Attention" (58 minutes) **404*

Cloud Environment Setup

Environment Setup	• get your UNC AI Gateway API key
	  ○ sent via Canvas Inbox
	• confirm access to the course models
	  ○ gpt-4.1-mini
	  ○ gpt-5.6-sol
	  ○ text-embedding-3-small
	• configure API keys and environment variables

First API Calls	• complete the provided Jupyter notebook • implement token counting and cost estimation • deploy basic LLM with security checks

<table>
<tr><td>Documentation</td><td>

document your setup process
note any challenges encountered

</td></tr>
<tr><td colspan="2">Live Session Preparation</td></tr>
<tr><td>Planned Discussion Items</td><td>

transformer architecture
challenges from the lab
bring questions about the readings

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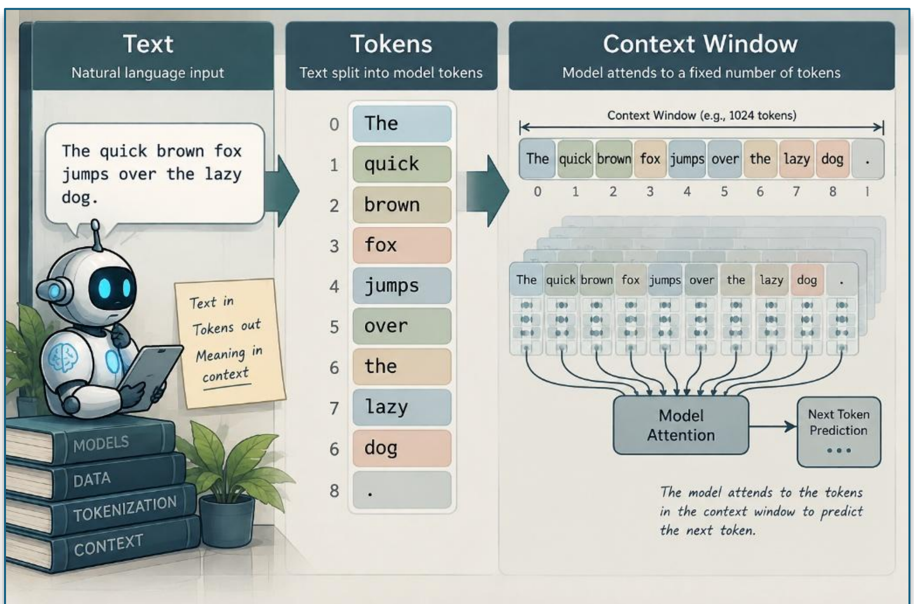
<tr><td>Key Concepts</td><td>

self-attention
how models attend to different parts of the input simultaneously
tokenization
how text is converted to numerical representations



ChatGPT 5.6 Sol

context window
maximum input size models can process
temperature
controls randomness in generation

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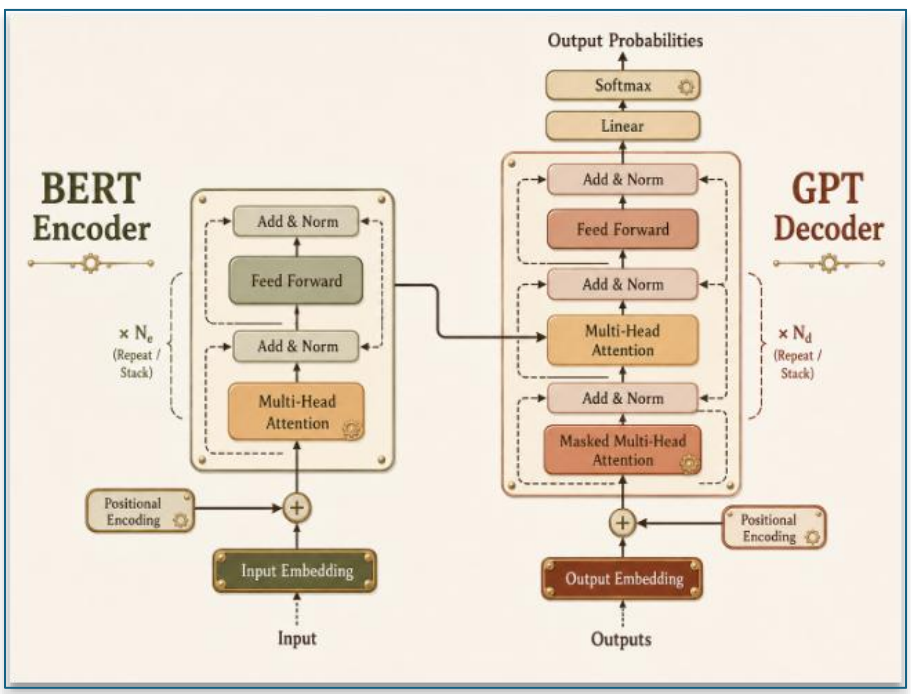
Async

Understanding Transformer Architecture

Encoder-Decoder Structure

- Encoder
    - a stack of identical layers
    - reads input sequence and generates a rich contextual representation
        - a set of context vectors

- Decoder
    - also a stack of layers
    - takes the encoders output
    - generates output sequence one token at a time
    - looks at what is already generated to inform the next word



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<td>Self-Attention Mechanism</td>
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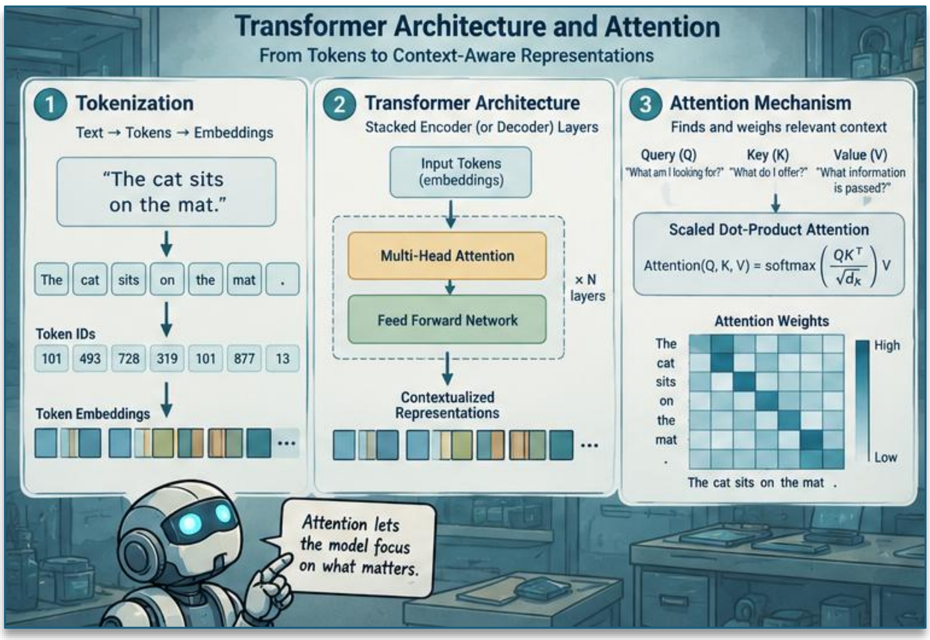
allows the model to look at all other words in the input sequence
provides added context for the current word
weighs the importance of all other words relative to a specific word
allows the model to capture complex, long-range dependencies and context



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<tr>
<td colspan="2">Understanding Attention Mechanisms</td>
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<tr>
<td>Query, Key, Value Vectors</td>
<td>

Query (Q)

the current word's question
"what am I looking for?"

Key (K)

the "label" of all words in the sequence
"what information do I hold?"

Value (V)

the actual content / meaning of the words

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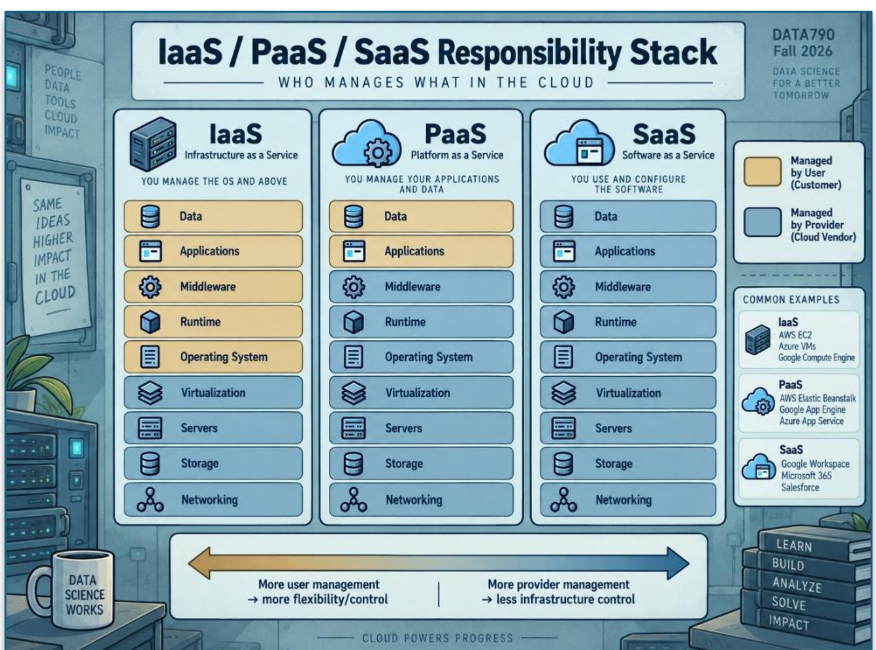
Score Calculation	• the mechanism calculates the score by matching the Q of each word with the K of another • similar to search engine matching your query (Q) string to a video title (K) to return a video (V)

Scaled Dot-Product Attention	• score
	  ○ compute the dot product of the Query with all Keys
	• scale
	  ○ divide scores by the square root of the key's dimension (d_k) to stabilize gradients
	• softmax
	  ○ normalizes all weights to sum to 1
	• output
	  ○ multiply the weights by the Value vectors and sum

Introduction to Cloud Platforms	
	

ChatGPT 5.6 Sol

<table>
<tr><td>IaaS</td><td>Infrastructure
delivers on-demand infrastructure
compute
storage
networking
user manages
OS
middleware
applications</td></tr>
<tr><td>PaaS</td><td>Platform
provides development platform
the cloud provider manages
underlying hardware
software resources</td></tr>
<tr><td>SaaS</td><td>Software
delivers a complete, cloud-based application
provider manages everything</td></tr>
<tr><td>Major Cloud Providers</td><td>Amazon Web Services (AWS)
the market leader
extensive and mature catalog of services
Microsoft Azure
strong integration with Microsoft’s Enterprise ecosystem
powerful hybrid cloud story
Google Cloud Platform (GCP)</td></tr>
</table>

	○ leader in AI, machine learning, and data analytics ○ leverages Google’s extensive infrastructure		
Model Landscape (video data is stale)			
Model	**GPT-4.0**	**Claude Sonnet 3.5**	**Gemini 1.5 Pro**
Capabilities	• fast all-rounder • native multimodal ○ text ○ audio ○ vision • ideal for real-time conversational AI	• reasoning specialist • leads in ○ top-tier reasoning ○ instruction following ○ coding benchmarks • excellent for complex tasks	• “Context King” • good multimodal skills • its defining feature is its massive context window for large-scale analysis
Context Windows	128K	200K	1M